

Torvik — continuous hydration sensing in a wearable band

The Problem

Dehydration measurably degrades cognitive and physical performance well before a person feels thirsty. There is no consumer device that measures hydration continuously; the clinical methods require a blood draw or a urine sample, neither of which anyone does four times a day.

Athletes, outdoor workers and older adults all manage hydration by guesswork.

Our Solution

Torvik is a wrist-worn band that measures tissue hydration continuously using bioimpedance spectroscopy across a frequency range we selected specifically for this, combined with skin temperature and sweat rate.

The engineering was hard and it is done. Wrist bioimpedance is noisy — motion artefacts, contact pressure, ambient temperature all corrupt the signal — and the last three years went into the electrode geometry and the signal chain that make the reading stable through movement.

We validated against plasma osmolality in a controlled study with 84 participants at a university exercise physiology department. Correlation was $r = 0.91$ across the hydration range tested, including during exercise. That study is under peer review.

The band runs nine days on a charge and is rated for swimming.

The Market

The wearables market is large, mature and dominated. We are not competing for it.

Our identified segments are endurance athletes, industrial workers in heat-exposed roles, and care settings for older adults where dehydration is a recognised cause of hospital admission.

We have not sized these segments with any rigour, and we should say so rather than present a number we made up.

Competition

No consumer device measures hydration. Two research groups have published on wrist bioimpedance for this purpose without productising it. Several smartwatches estimate hydration from heart rate and activity, which is an inference rather than a measurement.

We have the only validated continuous measurement we are aware of.

The Team

Dr. Henrik Sund spent nine years in biomedical sensing and holds three patents on the electrode design. Aiko Tanaka was a hardware engineer on two consumer wearables that shipped.

Twenty-two people, seventeen in hardware and signal processing.

Traction

- 400 units built on a pilot line
- validation study completed, $r = 0.91$ against plasma osmolality

- three patents granted
- 140 units placed with athletes and two industrial customers for field testing, at no charge

We have not sold a unit. We have not run a pre-order campaign. We do not have a signed purchase order from either industrial customer, and the athletes received theirs free.

We believe the demand is there. We have not demonstrated it.

Business Model

We intend to sell the band at \$249 with an optional subscription for the coaching layer. Bill of materials at pilot volume is \$118, which we expect to reach \$70 at 50,000 units.

Go-To-Market

To be determined. We have had conversations with two sports retailers and one occupational safety distributor.

Funding

We are raising \$12M to tool for production, complete regulatory clearance in two markets, and find out whether people buy this.